

READING PASSAGE 2

You should spend about 20 minutes on **Questions 14–26**, which are based on Reading Passage 2 on the following pages.

Questions 14–19

Reading Passage 2 has six paragraphs, **A–F**.

Choose the correct heading for each paragraph from the list of headings below.

Write the correct number, **i–viii**, in boxes 14–19 on your answer sheet.

List of Headings

- | | |
|-------------|--|
| i | Sending messages underground |
| ii | Potential advantages for farmers |
| iii | The widespread rejection of an idea |
| iv | One species accelerating growth in another |
| v | Some obstacles to plant communication |
| vi | Evidence leading to renewed belief in a theory |
| vii | New technologies which helped researchers |
| viii | A suggested benefit to the sender of plant communication |

14 Paragraph **A**

15 Paragraph **B**

16 Paragraph **C**

17 Paragraph **D**

18 Paragraph **E**

19 Paragraph **F**

How do plants talk to each other?

Research shows that plants communicate and interact with each other in surprisingly subtle and sophisticated ways.

- A** In 1983, plant scientists Jack Schultz and Ian Baldwin reported that the leaves of young maple trees increased their defense systems when exposed to maples that had been damaged by plant-eating (herbivorous) insects. The injured trees, they suggested, were alerting neighbors to the presence of a predator by releasing chemical signals into the air. But the plant research community did not accept this. The results were difficult to replicate, critics pointed out. Many also questioned how it could be evolutionarily stable if it benefited neighboring plants but not the plant releasing the signal. By the late 1980s, most ecologists felt that Schultz and Baldwin's ideas had been discredited.
- B** A decade later, however, a number of more carefully designed experiments began to yield convincing indications to the contrary. In 2000, evolutionary ecologist Richard Karban showed that wild tobacco plants became resistant to herbivores when grown in close proximity to sagebrush plants whose leaves had been damaged by cutting. This change appeared to be in response to chemicals – known as volatile organic compounds or VOCs – released by the sagebrush plants. Other researchers soon reported similar VOC-induced defense responses in several other plants, including lima bean, broad bean, barley, and corn. And in 2006, Karban showed that VOCs released by damaged sagebrush induce herbivore resistance in plants growing at distances of up to 60 cm, well within the range of sagebrush neighbors in nature.
- C** But the question still remains: 'Why should a plant waste valuable resources on a function which has no obvious advantage for it?' One hypothesis is that external communication channels are merely an extension of within-plant signaling. In sagebrush, lima bean, and poplar, VOCs released from damaged parts of a plant induce resistance in intact sections of the same plant, suggesting that each individual plant uses the signals to coordinate its own physiological responses to protect itself. Karban agrees, saying: 'The interplant signaling we see may be a result of plants co-opting that process.' Alternatively, VOC-based signaling between plants may have been favored because it enhances the 'extended fitness' of the sender by aiding related plants of the same species: a strategy known as kin selection.
- D** Over the past few years, a team led by Ariel Novoplansky of Ben-Gurion University of the Negev in Israel seems to have found proof that distress signals can be passed through the plants' roots. They planted garden pea plants in rows and subjected the first in each row to conditions similar to those experienced in a drought. They then evaluated the response by measuring the microscopic holes on leaves, known as pores, which react when there is a shortage of water. After fifteen minutes, the stressed plant was seen to be closing its pores, followed by all of its neighbors, one by one. Importantly, in a control setup where root contact between neighboring plants was

blocked, pores stayed open. Meanwhile, David Johnson's team at the University of Aberdeen in Scotland have been studying the labyrinths of hair-like fungi that occur around the roots of most plants. These fungi are involved in an important two-way relationship: in exchange for sugars, they provide plants with much-needed phosphorus and nitrogen. Research in which broad bean plants were infested with aphids – small herbivorous insects – revealed that these networks also served as a channel for warning neighboring plants of the infestation.

- E** Monica Gagliano of the University of Western Australia believes that plants may even use sounds to alert their neighbors. In one study, she demonstrated that chili plants growing next to fennel plants developed more quickly than seedlings grown with other chili plants. Gagliano and her colleagues suspect the chili plants were compensating for the presence of the fennel, which is known to release chemicals that inhibit the growth of other plants. Remarkably, however, when plant communication pathways – airborne volatiles, root contact, and common fungal networks – were blocked, the results begged for an alternative explanation. 'We think this other channel of communication might be acoustic,' says Gagliano. But behavioral ecologist Carel ten Cate of the University of Leiden in the Netherlands points out that taking advantage of such benefits would require sensory mechanisms yet to be described in plants. Despite widespread skepticism, Gagliano has received some encouragement. Richard Karban, for example, is cautiously enthusiastic. 'Whether or not the explanation she favors is the right one,' he says, 'I think that she's gotten results that, as a field, we need to come to grips with.'
- F** Researchers believe that knowledge of this phenomenon could eventually be applied to agriculture, and lead to the cultivation of hardier crops. But all agree that there is still much work to be done. 'Applying our limited knowledge [of plant-communication mechanisms] to agriculture is a big jump,' says Johnson, 'but it is definitely on the horizon.' Nevertheless, one message has emerged loud and clear from those studying this new realm of botanical interaction: despite not possessing eyes, ears, or a nervous system, plants are anything but uncommunicative. 'When I started my PhD (in the late 1980s), all this stuff was considered very weird,' recalls Ariel Novoplansky. 'Today, there is no doubt. We now recognise that plants are capable of some very sophisticated exchanges of information with other plants. This idea is not strange anymore.'

Questions 20–23

Complete the summary below.

Choose **ONE WORD ONLY** from the passage for each answer.

Write your answers in boxes 20–23 on your answer sheet.

Plant communication research in Israel and Scotland

Plants may send messages via their **20** _____ according to Ariel Novoplansky of Ben-Gurion University in Israel. In one study, Novoplansky's team planted rows of garden peas, and created conditions like a **21** _____ around the first plant in each row. The plants responded by closing the tiny holes on their leaf surfaces, known as pores. They then appeared to signal to other plants nearby, causing them to respond in a similar way.

Research at the University of Aberdeen in Scotland, meanwhile, has focused on the networks of **22** _____ which connect the plants beneath the soil. These structures, which extract sugars from the plants and supply them with phosphorus and nitrogen, also seem to play a key role in plant communication. Experiments involving broad bean plants and plant-eating insects called **23** _____ have also produced some interesting results.

Questions 23–26

Look at the following statements (Questions 23–26) and the list of researchers below.

Match each statement with the correct researcher, **A-E**.

Write the correct letter, **A-E**, in boxes 23–26 on your answer sheet.

- 24** It is likely that research into plant communication will one day help to improve our food supply.
- 25** Plant communication could occur as a consequence of internal mechanisms intended to help a single plant.
- 26** Attitudes towards plant communication have changed greatly in recent decades.

List of researchers

- A** Richard Karban
B Ariel Novoplansky
C David Johnson
D Monica Gagliano
E Carel ten Cate

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Questions 14–19 段落标题匹配

题号	答案	Heading 翻译	详细定位句	定位句翻译	详细解释
14	iii	一个观点被广泛否定	Paragraph A: “But the plant research community did not accept this... By the late 1980s, most ecologists felt that Schultz and Baldwin's ideas had been discredited.”	但是植物研究界并不接受这一点……到 20 世纪 80 年代末, 大多数生态学家认为 Schultz 和 Baldwin 的观点已经被否定了。	A 段先介绍 1983 年 Schultz 和 Baldwin 提出: 受伤植物会释放化学信号提醒邻近植物。但重点很快转向“这个观点不被接受”: 实验结果难以重复、批评者质疑其进化稳定性, 最后说这个观点被认为已经失信。因此本段核心不是“植物如何交流”, 而是早期理论被学界广泛否定。所以选 iii。
15	vi	证据使人们重新相信某一理论	Paragraph B: “A decade later, however, a number of more carefully designed experiments began to yield convincing indications to the contrary.”	然而, 十年后, 一系列设计更严谨的实验开始产生令人信服的反证。	B 段开头的“however”和“to the contrary”非常关键, 说明它在反转 A 段的否定态度。后文举 Karban 的实验: 野生烟草在受损 sagebrush 附近会增强抗虫能力, 其他植物也出现类似 VOC 诱导的防御反应。这些都是支持植物通过化学物质交流的证据。因此本段主旨是: 新的实验证据让人们重新相信植物交流理论, 对应 vi。
16	viii	植物交流对信号发送者的一个可能好处	Paragraph C: “Alternatively, VOC-based signalling between plants may have been favored because it enhances the 'extended fitness' of the sender by aiding related plants of the same species...”	另一种解释是, 植物之间基于 VOC 的信号传递之所以被保留下来, 可能是因为它通过帮助同物种的亲缘植物, 增强了信号发送者的“扩展适应度”。	C 段围绕一个问题展开: 植物为什么要浪费资源帮助别的植物? 其中一个解释是, 植物释放 VOC 本来是为了协调自身内部反应; 另一个解释是, 它能帮助同种亲缘植物, 从而提高发送者的“extended fitness”。题干 heading viii 中的 benefit to the sender 精准对应“enhances the extended fitness of the sender”。所以选 viii。
17	i	在地下发送信息	Paragraph D: “distress signals can be passed through the plants' roots”; “these networks also served as a channel for warning neighboring plants of the infestation.”	求救信号可以通过植物的根部传递; 这些网络也充当了警告邻近植物虫害的通道。	D 段主要讲地下通信: 一是豌豆通过根部传递干旱压力信号; 二是根部周围的真菌网络也能传递蚜虫入侵的警告。虽然段中提到 pores、fungi、phosphorus and nitrogen, 但这些都服务于“地下通道传递信息”这一主旨。因此选 i。
18	iv	一个物种促进另一个物种生长	Paragraph E: “chili plants growing next to fennel plants developed more quickly than seedlings grown with other chili plants.”	与茴香植物一起生长的辣椒植物, 比其他辣椒植物一起生长的幼苗发育得更快。	E 段开头讲 Gagliano 的研究: 辣椒在茴香旁边长得更快。这里就是 one species = fennel, another = chili plants, accelerating growth = developed more quickly 。后文虽然讨论声音交流、学界怀疑和 Karban 的评价, 但段落的核心例子是通过茴香和辣椒说明可能存在另一种交流方式。因此选 iv。
19	ii	对农民的潜在好处	Paragraph F: “Researchers believe that knowledge of this phenomenon could eventually be applied to agriculture, and lead to the cultivation of hardier crops.”	研究人员认为, 对这一现象的认识最终可以应用于农业, 并促成更强健作物的培育。	F 段开头直接说植物交流研究未来可能用于农业, 帮助培育更强健的作物。heading 中的“farmers”没有在原文直接出现, 但 agriculture 和 crops 都指向农业应用, 因此是同义概括。后文虽然说仍有很多工作要做, 但主旨仍是这一发现未来可能有农业价值。因此选 ii。

Questions 20–23 填空题

题号	答案	题干翻译	详细定位句	定位句翻译	详细解释
20	roots	根据以色列本-古里安大学的 Ariel Novoplansky 的研究, 植物可能通过它们的_____发送信息。	Paragraph D: “a team led by Ariel Novoplansky... seems to have found proof that distress signals can be passed through the plants’ roots.”	Ariel Novoplansky 领导的团队似乎已经找到证据, 证明求救信号可以通过植物的根部传递。	题干中的 “send messages via their...” 对应原文 “distress signals can be passed through...”。 “via” 就是 “through” 的同义替换。空格前有 their, 所以填复数名词 roots 。
21	drought	在一项研究中, Novoplansky 的团队种植了一排排豌豆, 并在每排第一株植物周围制造了类似_____的条件。	Paragraph D: “They planted garden pea plants in rows and subjected the first in each row to conditions similar to those experienced in a drought.”	他们把花园豌豆排成行种植, 并让每一排的第一株植物处于类似干旱时所经历的条件中。	题干中的 “created conditions like...” 对应原文 “subjected... to conditions similar to...”。空格要填的是这种条件类似什么, 原文明确是 a drought 。注意答案只能填一个词, 所以填 drought , 不要写 “a drought”。
22	fungi	与此同时, 苏格兰阿伯丁大学的研究关注的是_____网络, 这些网络在土壤下连接植物。	Paragraph D: “David Johnson’s team... have been studying the labyrinths of hair-like fungi that occur around the roots of most plants.”	David Johnson 的团队一直在研究大多数植物根部周围出现的、迷宫般的毛发状真菌。	题干中的 “networks of _____ which connect the plants beneath the soil” 对应原文后文 “these networks also served as a channel...”。原文说这些网络是由根部周围的 hair-like fungi 构成的。因此填 fungi 。不能填 “networks”, 因为题干已经有 networks of。
23	aphids	涉及豌豆植物和被称为_____的食植物昆虫的实验也产生了一些有趣结果。	Paragraph D: “Research in which broad bean plants were infested with aphids – small herbivorous insects – revealed that these networks also served as a channel for warning neighboring plants of the infestation.”	一项让豌豆植物遭受蚜虫侵害的研究显示, 这些网络也充当了警告邻近植物遭受虫害的通道; 蚜虫是一种小型食草昆虫。	题干中的 “plant-eating insects called...” 对应原文插入解释 “aphids – small herbivorous insects”。herbivorous insects = plant-eating insects。因此填 aphids 。注意 aphids 是复数形式, 应保留原文形式。

Questions 24–26 研究者配对题

题号	答案	题干翻译	详细定位句	定位句翻译	详细解释
24	C	植物交流研究很可能有一天会帮助改善我们的食物供应。	Paragraph F: “Applying our limited knowledge [of plant-communication mechanisms] to agriculture is a big jump,’ says Johnson, ‘but it is definitely on the horizon.’”	Johnson 说: “把我们对植物交流机制的有限认识应用到农业中是一个很大的跨越, 但这件事肯定已经在视野中了。”	题干中的 “will one day help to improve our food supply” 对应原文的 “applied to agriculture” 和 “cultivation of hardier crops”。真正说这句话的人是 Johnson , 也就是 David Johnson, 虽然 F 段开头是 “Researchers believe...”, 但后面用引号具体表达农业应用前景的是 Johnson, 因此选 C 。
25	A	植物交流可能是帮助单个植物的内部机制所产生的结果。	Paragraph C: “One hypothesis is that external communication channels are merely an extension of within-plant signaling... Karban agrees, saying: ‘The interplant signaling we see may be a result of plants co-opting that process.’”	一种假说是, 外部交流通道只不过是植物内部信号传递的延伸.....Karbon 同意这一点, 他说: “我们看到的植物间信号传递, 可能是植物借用了这个过程的结果。”	题干中的 “internal mechanisms intended to help a single plant” 对应原文 “within-plant signaling” 和 “each individual plant uses the signals to coordinate its own physiological responses to protect itself”。也就是说, 这种机制本来是单株植物内部用来保护自己的, 后来可能被植物间交流 “借用”。这句话由 Karban 表示同意并解释, 所以选 A 。
26	B	近几十年来, 人们对植物交流的态度发生了巨大变化。	Paragraph F: “When I started my PhD (in the late 1980s), all this stuff was considered very weird,’ recalls Ariel Novoplansky. ‘Today, there is no doubt... This idea is not strange anymore.’”	Ariel Novoplansky 回忆说: “我在 20 世纪 80 年代末开始读博士时, 这些东西都被认为非常奇怪。如今, 这一点已经毫无疑问.....这个想法已经不再奇怪了。”	题干的 “attitudes... have changed greatly in recent decades” 对应原文从 “very weird” 到 “there is no doubt / not strange anymore” 的变化。说这句话的人是 Ariel Novoplansky, 因此选 B 。